

TIANLIN LIU

✉ t.liu@unibas.ch 🏠 tianlinliu.com 🌐 liutianlin0121

Updated: 19th September 2023

EDUCATION

- | | |
|--------------|--|
| 2020–PRESENT | Ph.D. Computer Science
University of Basel, Switzerland
Advisor: Ivan Dokmanić |
| 2017–2019 | M.S. Data Engineering
Jacobs University Bremen, Germany
Advisor: Herbert Jaeger |
| 2013–2016 | B.S. Mathematics
Jacobs University Bremen, Germany
Advisor: Götz Pfander |

WORK EXPERIENCE

- | | |
|-------------|---|
| Fall 2023 | Research Scientist intern at Google DeepMind, Paris.
Aligning language models with RL-free methods. Hosted by Mathieu Blondel |
| Summer 2022 | Research Scientist intern at Google Brain, Zurich.
Improved vision mixture-of-experts models via optimal transport. Hosted by Joan Puigcerver and Mathieu Blondel; collaborated with Carlos Riquelme. |

AWARDS AND HONORS

- | | |
|------|---|
| 2023 | Google Conference Scholarship |
| 2022 | Young Researcher, Heidelberg Laureate Forum. |
| 2022 | Travel Award, SIAM Conference on Imaging Science. |
| 2021 | Oberwolfach Leibniz Graduate Student Grant, Oberwolfach Institute. |
| 2019 | Dean's Award for Outstanding Master's Thesis, Jacobs University Bremen. |
| 2019 | IEEE NER 2019 Best Paper Finalist Award. |
| 2018 | SmartStart-1 Fellowship, Bernstein Network and Volkswagen Foundation. |

PUBLICATIONS

- [9] [Tianlin Liu](#), Joan Puigcerver, and Mathieu Blondel. Sparsity-Constrained Optimal Transport. In Proceedings of the 11th International Conference on Learning Representations (ICLR 2023). [\[PDF\]](#)
Spotlight presentation
- [8] [Tianlin Liu](#), Anadi Chaman, David Belius, and Ivan Dokmanić. Learning Multiscale Convolutional Dictionaries for Image Reconstruction. IEEE Transactions on Computational Imaging, 2022. [\[PDF\]](#)
- [7] Anastasis Kratsios, Behnoosh Zamanlooy, [Tianlin Liu](#), and Ivan Dokmanić. Universal Approximation Under Constraints is Possible with Transformers. In Proceedings of the 10th International Conference on Learning Representations (ICLR 2022). [\[PDF\]](#)
Spotlight presentation
- [6] [Tianlin Liu](#) and Friedemann Zenke. Finding Trainable Sparse Networks through Neural Tangent Transfer. In Proceedings of 37th International Conference on Machine Learning (ICML 2020). [\[PDF\]](#)
- [5] Zekun Yang* and [Tianlin Liu](#)* (*Equal contribution). Causally Denoise Word Embeddings Using Half-Sibling Regression. In Proceedings of 34th AAAI Conference on Artificial Intelligence (AAAI 2020). [\[PDF\]](#)
- [4] [Tianlin Liu](#). Harnessing Slow Dynamics in Neuromorphic Computation. Master’s thesis, 2019. [\[PDF\]](#)
Dean’s Prize for Outstanding Master’s Thesis
- [3] [Tianlin Liu](#), Lyle Ungar, and João Sedoc. Continual Learning for Sentence Representations Using Conceptors. [\[PDF\]](#)
In Proceedings of the 17th Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2019).
- [2] Xu He, [Tianlin Liu](#), Fatemeh Hadaeghi, and Herbert Jaeger. Reservoir Transfer on Analog Neuromorphic Hardware. In Proceedings of the 9th International IEEE Conference on Neural Engineering (NER 2019). [\[PDF\]](#)
Best Paper Finalist Award (3rd place out of 467 accepted papers)
- [1] [Tianlin Liu](#), Lyle Ungar, and João Sedoc. Unsupervised Post-processing of Word Vectors via Conceptor Negation. In Proceedings of 33th AAAI Conference on Artificial Intelligence (AAAI 2019). [\[PDF\]](#)

OPEN-SOURCE CONTRIBUTION

huggingface/transformers : the go-to library for transformers	☆ 111k
huggingface/trl : lightweight library for RLHF	☆ 5.3k
PythonOT/POT : python optimal transport	☆ 2k
google-research/vmoe : vision mixture of experts models	☆ 336
google/jaxopt : differentiable optimizers in JAX	☆ 735

JOURNAL AND CONFERENCE REVIEWING

CONFERENCE	ICML (2021–2023), NeurIPS (2021–2023), ICLR (2022, 2023), LoG (2022)
JOURNAL	IEEE Transactions on Signal Processing, IEEE Signal Processing Letters, IEEE Open Journal of Signal Processing

TEACHING EXPERIENCE

2022	Mathematics of Data Science	Uni Basel
	Scientific Writing	Uni Basel
	Machine Learning on Graphs, Groups, and Manifolds	Uni Basel
2021	Computational Sustainability and AI for Social Good	Uni Basel
2021	High-dimensional Data Analysis and Learning on Graphs	Uni Basel
2018, 2019	Machine Learning	Jacobs Uni Bremen
2018, 2019	Principles of Statistical Modeling	Jacobs Uni Bremen